

PRIYADARSHINI COLLEGE OF ENGINEERING, NAGPUR

Department: Computer Technology

Semester: VIII Section: A and B

CAT-I (2023-24)

Subject: Blockchain and its Applications (Elective VI)I

Subject Code: BECT802T-2

Duration: 1.5Hrs

Max. Marks: 35

Note:

- 1) All questions are compulsory.
- 2) All questions carry marks as indicated.
- 3) Due credits will be given on neatness.
- 4) Draw diagram wherever it is necessary.

Questions		Marks	CO	BL
Q.1	A. Which of the following is used to point to a block in blockchain a. Hash Pointer b. User Id c. Transaction ID d. Timestamp.	1	CO1	1
	B. What are the features of a Hash Function? a. Puzzle Friendly b. Collision Resistance c. Deterministic d. Post Image resistance	1	CO1	1
	C. Alice employs the RSA cryptosystem with the prime numbers $P=7$ and $Q=17$ to derive her public and private keys. Given that Alice's public key is 11, what will be her private key?	5	CO1	3
	D. Discuss SHA256 Algorithm with processing.	7	CO1	2
OR				
Q.2	A. For SHA 256 bit hash function, the attacker needs to compute how many hash operations in order to find two matching outputs. a. 0.3×2^{128} b. 0.2×10^{50} c. 0.25×2^{130} d. 1×2^{256}	1	CO1	3
	B. Where are the ledger logs stored in a blockchain? a. On a SQL database b. On a central immutable ledger c. On a meta data table d. In ledger of each pair	1	CO1	2
	C. What is Hash Pointer? Discuss tamper detection using hash chain	5	CO1	2
	D. What is Digital Signature? Explain	7	CO1	2
Q.3	A. In the Bitcoin block header, the block identifier is calculated a. Using SHA256 on the current block header b. Using Double SHA256 on the previous block hash c. Using Double SHA256 on the Difficulty bits d. Using Double SHA256 on the current block header	1	CO2	1
	B. What is nonce? a. The transaction ID number b. The number miners run through to generate a correct hash c. The generator point used in elliptic curve cryptography d. A miner ASIC chip array	1	CO2	1
	C. What are smart Contracts? Explain	4	CO2	2

	D. Define Blockchain. Explain structure of Block and block header.	8	CO2	2
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OR

Q.4	A. Which of these fields is present in a Bitcoin block summary? a. Nonce b. Gas Used c. Gas Limit d. Private Key of the Sender	1	CO2	1
	B. DLT can be used to maintain financial information only. a. False b. True	1	CO2	1
	C. What is double Spending? Explain how double spending is handled using blockchain.	5	CO2	2
	D. What are the steps for joining a bitcoin network? Explain.	7	CO2	2
Q.5	A. Permissioned blockchain is regarded as more secure than open blockchain as the participants are known beforehand and pre-authenticated. a. True b. False	1	CO3	1
	B. What are some Bitcoin exchanges available in India: Please select the most appropriate choice among the options. a. BuyUCoin b. ZebPay c. WazirX i. a and b ii. b and c iii. a and c iv. a, b and c	1	CO3	1
	C. Discuss I. Proof of Work (PoW) II. Proof of Stack	5	CO3	2

OR

Q.6	A. What is the correct order of adding a new block to blockchain i. Block Mining ii. Block propagation iii. Block Flooding iv. Transaction Flooding a. iii, iv, ii, i b. iv, i, iii, ii c. iv, iii, ii, i d. ii, i, iii, iv	1	CO3	1
	B. Double spending is reusing digital assets intentionally or inadvertently. True or False? a. True b. False	1	CO3	1
	C. Differentiate Permissioned and Permission less Blockchain	5	CO3	2